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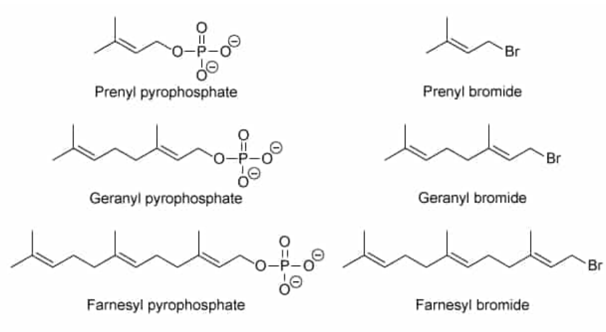

Figure 1 Structures of prenyl pyrophosphates and prenyl bromides

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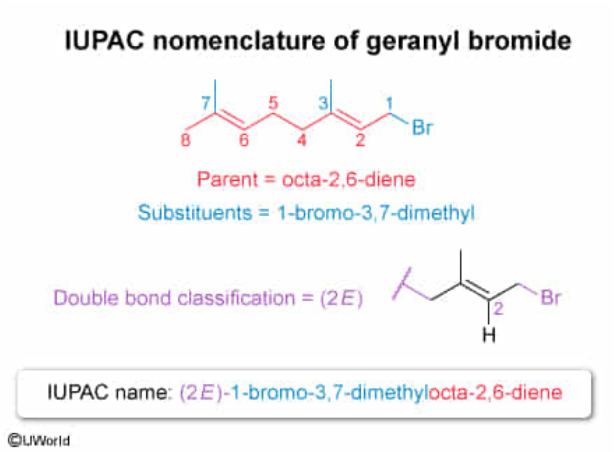
In these molecules, the pyrophosphate moiety serves as a leaving group and activates the allylic system for nucleophilic attack by protein residues. In this way, allylic pyrophosphates serve a similar biosynthetic purpose to allylic halides (eg, prenyl bromide, geranyl bromide, farnesyl bromide) that are traditionally used in synthetic substitution reactions. Activated allylic electrophiles are amenable to substitution reactions that proceed through either a first-order (S_N1) or second-order (S_N2) mechanism.

What is the IUPAC name of geranyl bromide?

- A. (6*E*)-8-bromo-2,6-dimethylocta-2,6-diene [14%]
- B. (7*E*)-8-bromo-2,6-dimethylocta-3,7-diene [2%]
- C. (2*E*)-1-bromo-3,7,7-trimethylhepta-2,6-diene [4%]
- D. (2*E*)-1-bromo-3,7-dimethylocta-2,6-diene [78%]

Correct
78% answered correctly

Explanation:



IUPAC provides a systematic method of naming organic compounds. A compound's **parent chain** is the longest continuous chain of carbon atoms that contains the greatest number of highest-priority functional groups, and it is numbered to provide the highest-priority functional groups with the lowest possible number. The parent chain name includes a prefix denoting the number of carbon atoms in the chain, a middle portion denoting the presence of select pi-bond functional groups, and a suffix denoting the quantity and type of highest-priority functional group.

Lower-priority functional groups are named as substituents, which include a numerical position of attachment to the parent chain. Substituent chains are numbered from the point of attachment to the parent chain and named using a prefix to indicate the carbon chain length and a –*yl* suffix. Substituents are listed in alphabetical order, followed by the parent chain.

Stereocenters within a molecule are listed at the beginning of the compound name in parenthesis using the number of the parent chain to identify its position and are classified as *R*/*S* or *E*/*Z*.

The parent chain in geranyl bromide contains eight carbon atoms (*oct*– prefix) and two double bond functional groups (–*diene* suffix). Since the parent chain can be numbered two ways, the correct solution numbers the parent chain to provide the lowest number for the first substituent: octa-2,6-diene. The double bond starting on carbon 2 is a stereocenter and is assigned an *E* configuration, and substituents are located at positions 1 (bromo), 3, and 7 (dimethyl). Therefore, the IUPAC name of geranyl bromide is **(2*E*)-1-bromo-3,7-dimethylocta-2,6-diene**.

(Choice A) The parent chain is numbered to provide substituents with the lowest possible numbering. (6*E*)-8-bromo-2,6-dimethylocta-2,6-diene incorrectly numbers the parent chain.

(Choice B) The position of each double bond is indicated by the *lowest* number of the two carbons that participate in the alkene. (7*E*)-8-bromo-2,6-dimethylocta-3,7-

diene identifies the double bond locations using the highest rather than lowest position numbers.

(Choice C) (2*E*)-1-bromo-3,7,7-trimethylhepta-2,6-diene incorrectly numbers the parent chain and includes an additional methyl substituent at carbon 7 instead of including the methyl in the parent chain.

Educational objective:
IUPAC provides a systematic method to generate a name for organic compounds. The parent chain is numbered to give substituents the lowest numerical position.

Time spent: 0 sec
QID:403769
Subject:
Organic Chemistry
System:
Introduction to Organic Chemistry

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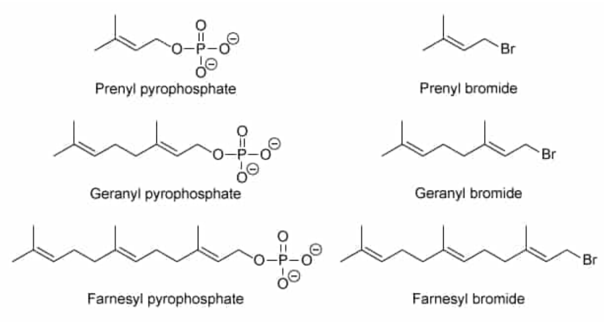

Figure 1 Structures of prenyl pyrophosphates and prenyl bromides

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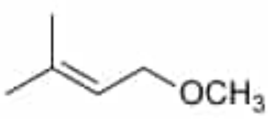
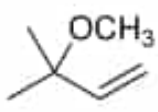
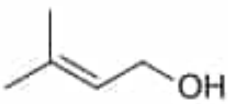
If prenyl bromide is heated with methanol, what will be the structures of the substitution products that are formed?



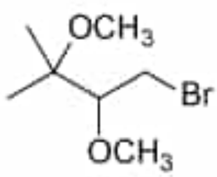
I



II



III



IV

- A. I only [31%]
- B. I and II only [45%]
- C. I and III only [17%]
- D. II and IV only [5%]

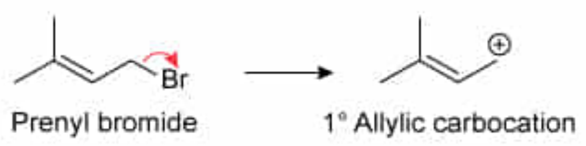
Correct
44% answered correctly

Explanation:

In a **substitution reaction**, a functional group in the **substrate** is displaced by a new functional group. The lost group is called the leaving group (LG) and serves to make the carbon electrophilic leave with the electrons in the C-LG sigma bond. The new functional group is the nucleophile, and it creates a new sigma bond by donating one of its lone pairs of electrons. Substitution reactions can be either first order (S_N1) or second order (S_N2).

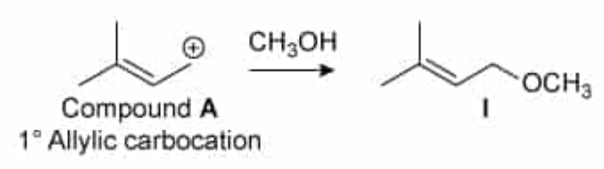
S_N1 reactions proceed through a planar carbocation intermediate, which is formed when the leaving group fully ionizes the C-LG bond. The nucleophile then attacks the electrophilic carbocation and creates the new sigma bond. S_N1 reactions are susceptible to carbocation rearrangement. Carbocation formation is the rate-limiting step (ie, slowest step) in an S_N1 reaction, which is favored for weaker nucleophiles (ie, substrates that can stabilize the carbocation intermediate) and heated reactions.

Prenyl bromide is classified as an allylic substrate that favors either S_N1 or S_N2 reactions. Since methanol is a weak nucleophile and the reaction requires heat, an S_N1 reaction mechanism is favored. Bromide acts as the LG, and a primary allylic carbocation forms when bromide leaves.

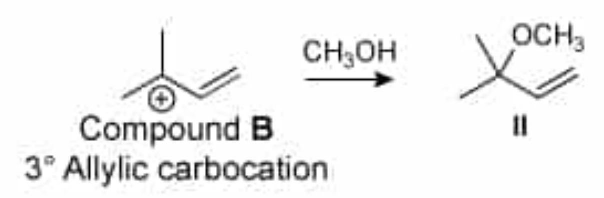


Because the allylic carbocation has two major resonance forms, methanol can attack *either* resonance form to generate a mixture of two products.

Nucleophilic attack on resonance form A leads to methanol addition on a primary carbocation **(Number I)**:



whereas nucleophilic attack on resonance form B leads to methanol addition on a tertiary carbocation **(Number II)**.



(Number III) For this alcohol-containing product to be formed through an S_N1 reaction, the nucleophile would need to be water, not methanol.

(Number IV) This compound results from the addition of two methanol molecules to the double bond in prenyl bromide, which is *not* a substitution reaction.

Educational objective:
Substitution reactions can be either first order or second order and can be differentiated by analyzing the features of the electrophilic substrate, nucleophile, and reaction conditions. S_N1 reactions that generate an allylic carbocation can lead to the formation of multiple substitution products.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403770
System:
Introduction to Organic Chemistry

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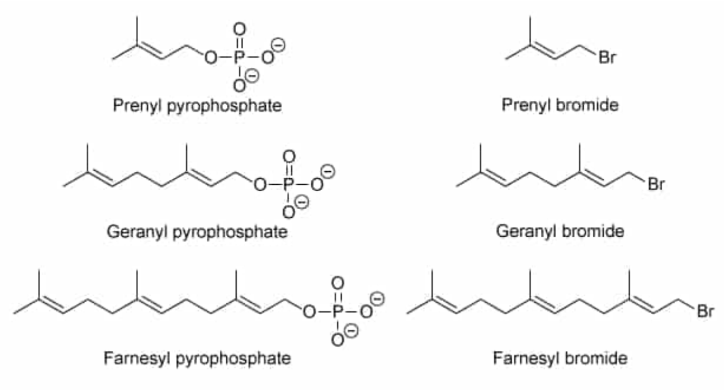

Figure 1 Structures of prenyl pyrophosphates and prenyl bromides

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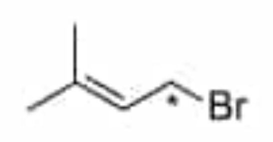
When prenyl bromide undergoes an S_N1 substitution with methanol, what hybridization state changes does the carbon atom marked with an asterisk undergo?



A. $sp^2 \rightarrow sp^3 \rightarrow sp^2$ only [7%]

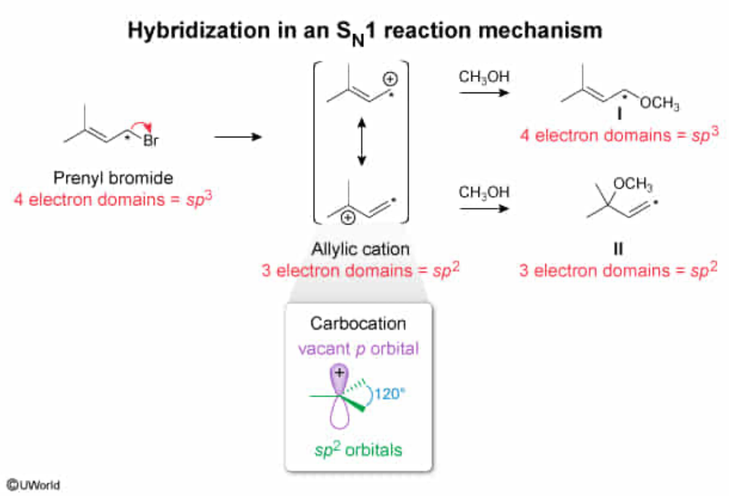
B. $sp^3 \rightarrow sp^2 \rightarrow sp^3$ only [40%]

C. $sp^3 \rightarrow sp^2 \rightarrow sp^2$ only [14%]

D. Both $sp^3 \rightarrow sp^2 \rightarrow sp^3$ and $sp^3 \rightarrow sp^2 \rightarrow sp^2$ [37%]

Correct
36% answered correctly

Explanation:



In discrete atoms, electrons are located in atomic orbitals that describe the probability clouds of electron density in space surrounding the atom's nucleus. Atomic orbitals have different **shapes**, including *s* orbitals, *p* orbitals, *d* orbitals, and *f* orbitals. Hybridization is the mathematical recombination of atomic orbitals to produce new orbitals bearing features of the contributing orbitals. The process of hybridization demonstrates conservation of orbitals.

Orbital hybridization is integrally related to the number of electron domains surrounding a nucleus and the resulting molecular geometry. The number of electron domains is determined by the sum of all sigma bonds and nonbonding electron environments. Vacant orbitals are not included in an electron domain total (eg, atoms with two electron domains have an *sp* hybridization state with two unhybridized *p* orbitals).

In prenyl bromide, the indicated carbon atom contains four sigma bonds, and therefore four electron domains:

- the adjacent carbon atom
- the bromine atom
- the two hydrogen atoms

Consequently, the indicated carbon's hybridization state is sp^3. When the bromide ion is lost, a carbocation with three sigma bonds and a vacant *p* orbital (ie, an sp^2 hybridization state) is formed. Since the carbocation is allylic and has two major resonance forms, the methanol nucleophile can attack either resonance form and leads to the possibility of two major substitution products: one with four electron domains and an sp^3 hybridization state, and another with three electron domains and an sp^2 hybridization state.

Therefore, the indicated carbon undergoes **both $sp^3 \rightarrow sp^2 \rightarrow sp^3$ and $sp^3 \rightarrow sp^2 \rightarrow sp^2$** hybridization changes.

(Choice A) The substrates and products of an S_N1 reaction generally have an sp^3 hybridization state, with an intermediate carbocation that has an sp^2 hybridization state. This choice is the reverse pattern of hybridization changes.

(Choices B and C) Although each of these hybridization changes are correct for one of the two major substitution reaction products, *both* hybridization changes occur, not just one.

Educational objective:

Atomic hybridization can be determined by the number of electron domains, which may change throughout a reaction process.

Time spent: 0 sec	**QID**:403771
Subject: Organic Chemistry	**System:** Introduction to Organic Chemistry

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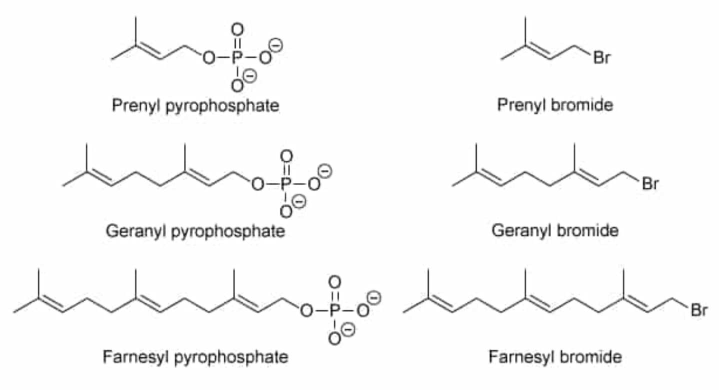

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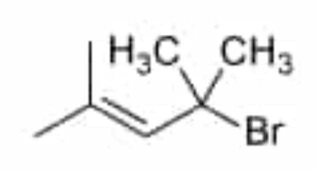
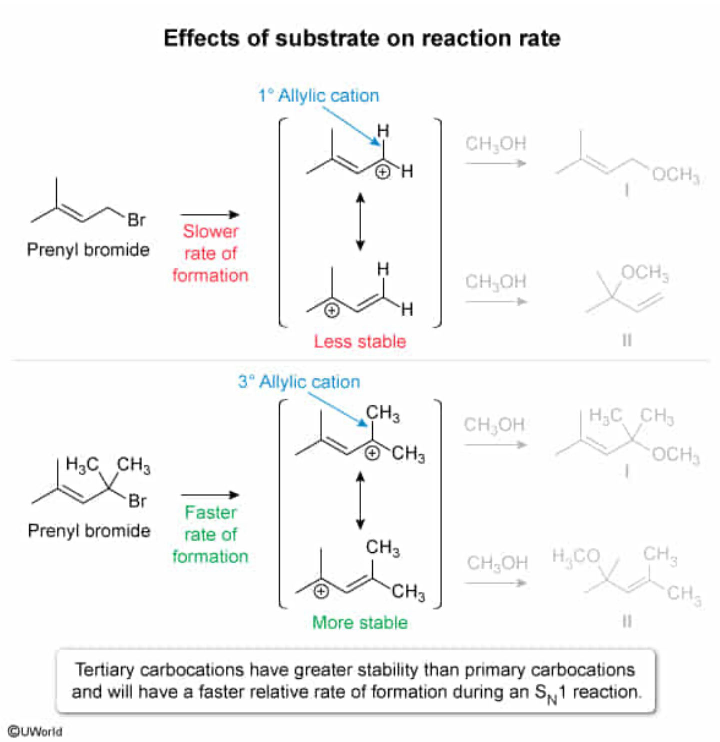
If the molecule shown below replaces prenyl bromide, what would be the impact on its rate of S_N1 substitution reaction with methanol?



- ✔ A. The rate would increase. [52%]
- B. The rate would decrease. [39%]
- C. The rate would not change. [6%]
- D. Nothing can be inferred about the rate of reaction. [1%]

Correct
51% answered correctly

Explanation:



In an S_N1 reaction, the slowest, **rate-limiting step** is the **formation of** the initial **carbocation**. Consequently, molecular features that impact the **carbocation stability** directly impact both the **rate** of carbocation formation and the **overall rate** of the S_N1 reaction.

Since they have only six valence electrons, all carbocations are electron deficient. The inclusion of alkyl groups is one of the primary ways to provide additional stability for a carbocation. Adjacent alkyl groups donate electron density along their sigma bonds as a manifestation of the inductive effect. Alkyl group stabilization of carbocations is additive, where methyl carbocations have the least stability and tertiary carbocations have the greatest stability.

In addition, adjacent alkyl groups stabilize the electron-deficient *p* orbital of a carbocation through a process called hyperconjugation, where alkyl group sp^3 orbitals provide secondary overlap with the unhybridized *p* orbital. Carbocations with higher stability have an inherently lower relative energy and, by the Arrhenius equation, a greater rate of formation.

Prenyl bromide is a primary alkyl halide and forms a primary carbocation when bromine leaves. The replacement molecule for prenyl bromide contains two additional methyl groups at the carbon atom bearing the bromine leaving group (ie, a tertiary alkyl halide). Consequently, the resultant carbocation will be tertiary and contain two additional alkyl groups relative to prenyl bromide. Because **tertiary carbocations** are **more stable** than primary carbocations, the alternate carbocation will be more stable and **form more quickly**. Therefore, the **rate** of the S_N1 reaction would be expected to **increase**.

(Choices B and C) The alternate substrate is a tertiary alkyl halide and would form a more stable tertiary carbocation during an S_N1 reaction. As a result, the rate would be expected to increase.

(Choice D) The structure of the alternate substrate is sufficient to determine that the substrate was changed from a primary alkyl halide to a tertiary alkyl halide, which are better substrates for an S_N1 reaction.

Educational objective:
The rate-limiting step in an S_N1 reaction is the formation of the initial carbocation. Molecular features that impact carbocation stability have

an impact on the rate of S_N1 reactions. Tertiary carbocations have the greatest stability and provide the highest rate of S_N1 reactions.

Time spent: 0 sec	**QID**:403772
Subject:	**System:**
Organic Chemistry	Introduction to Organic Chemistry

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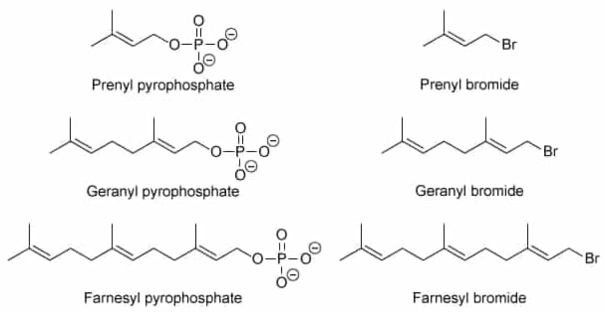

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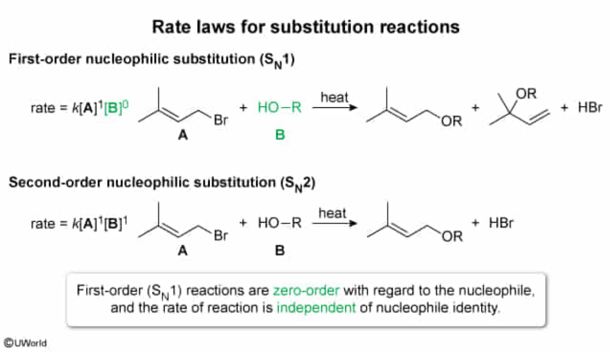
For the heated reaction of prenyl bromide with methanol, if the methanol is replaced with a different alcohol, which alcohol would provide the slowest rate of substitution reaction?

- A. 2-butanol [5%]
- B. 2-methyl-2-propanol [67%]
- C. 1-butanol [11%]
- ✔ D. The identity of the alcohol does not impact the rate of the described substitution reaction. [16%]

Correct

16% answered correctly

Explanation:



The structural features of both the **nucleophile** and the **electrophile** have the potential of impacting the rate of a **substitution reaction**. The steric hinderance of a nucleophile can impact the rate of a substitution reaction. Linear, unhindered nucleophiles are more capable of approaching the electrophilic carbon atom with the atomic/hybrid orbital containing the nucleophilic electron pair whereas more-substituted, sterically hindered nucleophiles have reduced capability of approaching the electrophilic carbon atom.

This question is about evaluating the rate impacts of replacements for the nucleophile (alcohol), where the identity of the electrophile (prenyl bromide) remains constant. Prenyl bromide is an allylic halide and can participate in both first order (S_N1) and second order (S_N2) substitution reactions.

The substitution reaction described in this question involves a weak nucleophile (methanol) and heat. Together, these factors suggest that the substitution reaction will proceed through a first order (S_N1) mechanism. The rate-limiting step in an S_N1 reaction is the formation of a **carbocation intermediate**. As such, the rate law for an **S_N1 reaction** is **first order** with respect to the **electrophile** (prenyl bromide) and **zero order** with respect to the **nucleophile** (methanol). The identity and concentration of a zero-order reactant are rate-independent.

The most-substituted, tertiary alcohol 2-methyl-2-propanol has the greatest amount of **steric hinderance** and would otherwise be expected to provide the slowest rate of substitution product formation. However, since the described reaction is a first order (S_N1) reaction that is zero order with respect to the nucleophile (alcohol), the **identity of the alcohol will not impact the rate** of substitution product formation.

(Choices A, B, and C) The classification of alcohol choices as methyl, primary, secondary, or tertiary will not impact the rate of the substitution reaction in an S_N1

mechanism, as the rate law for this process is zero order with respect to the nucleophile (alcohol).

Educational objective:
The rate of substitution reactions can be impacted by structural features of either the nucleophile or the electrophile. First-order (S_N1) substitution reactions are zero order with respect to the nucleophile, and the identity of the nucleophile will not affect the rate of S_N1 product formation.

Time spent: 0 sec
Subject: Organic Chemistry

QID: 403773
System: Introduction to Organic Chemistry